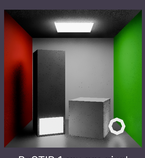
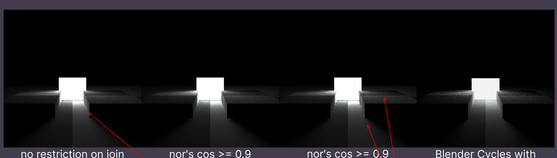
ReSTIR DI



ReSTIR 1 ray per pixel
BDRF: Lambert + GGX (different roughness)
Init samples 5 x MIS(Area Light Importance, GGX)
Temporal accumulation: 4 frames
Spatial accumulation: 12 samples

ReSTIR Lambert spatial join study

BDRF: Lambert
Init samples 5 x Area Light Importance
Temporal accumulation: 2 frames
Spatial accumulation: 5 samples radius 10



no restriction on join

nor's cos >= 0.9
fireflies on box edge
removed

nor's cos >= 0.9
pos's dist < 0.04
better shadow
lower convergence

Blender Cycles with
10 rays per pixel, no
denoise, no indirect

ReSTIR GGX spatial join study

BDRF: GGX (different roughness)
Init samples 5 x MIS(Area Light Importance, GGX)
Temporal accumulation: 2 frames
Spatial accumulation: 5 samples radius 10

nor's cos < 0.9
pos's dist < 0.04
Specular over spreads

nor's cos >= 0.9
pos's dist < 0.04
half's cos >= 0.9
Adding restriction between
half vector's norrows the
specular

nor's cos >= 0.9
pos's dist < 0.04
half's cos >= 0.98
better specular
lower convergence

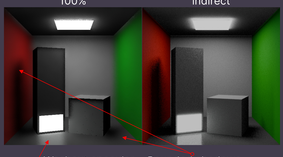
Blender Cycles with
10 rays per pixel, no
denoise, no indirect

Average numbers of samples merged per pixel using under different join restrictions:
73.51 [100%] - no restriction on spatial join
57.97 [78%] - pos's dist < 0.1
35.07 [47%] - pos's dist < 0.04 [good shadows]
33.12 [45%] - pos's dist < 0.1 and nor's cos >= 0.9 [OK shadows, good lambert]
26.35 [35%] - pos's dist < 0.1 and nor's cos >= 0.9 and half's cos >= 0.98 [ok shadows, good lambert, good ggx]
49.41 [67%] - half's cos >= 0.98

Lambert + GGX, 5 x MIS(Area Lights Importance, GGX), Temporal 4, Spatial 12 from radius 10
Visual impact & reservoir join percentage vs join restriction

Start point
No restriction on spatial.
Average join: 13583.87
100%

Visual reference
Blender Cycles with 10
samples, no denoise, no
indirect



Washout specular + Rounded shadows
+ Dark visual artefacts

half's cos >= 0.98
average join: 6768.61
49.82%
+ Good specular

half's cos >= 0.98
nor's cos > 0.9
average join: 6478.03
47.68%

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.5
average join: 6469.80
47.62%
+ No artefacts

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.1
average join: 3692.08
27.17%
+ OK, shadows

half's cos >= 0.98
nor's cos > 0.9
pos's dist < 0.04
average join: 190.19
1.40%
+ Good Shadows

- Artefacts: nor's cos >= 0.9 and pos's dist < 0.5 is needed to remove artefacts

- Lambert: it looks good everywhere (once the artefacts are removed)

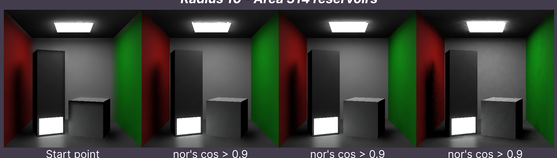
- GGX: half's cos >= 0.98 results in good specular but we drop half the spatial samples

- Shadows: best shadows with pos's dist < 0.04 (worse joining 1.4%). Acceptable shadows begins from pos's dist < 0.1, 27.17%

A lower percentage of considered neighbors joining means that convergence will require more frames.

Lambert + GGX, 5 x MIS(Area Lights Importance, GGX), Temporal 4, Spatial 12 samples
no half's restriction

Radius 10 - Area 314 reservoirs



Start point
no restriction on spatial
average join: 13583.87
100%

nor's cos > 0.9
pos's dist < 0.5
average join: 12783.13
47.62%

nor's cos > 0.9
pos's dist < 0.1
average join: 6833.63
27.17%

nor's cos > 0.9
pos's dist < 0.04
average join: 334.35
1.40%

Radius 5 - Area 78.5 reservoirs

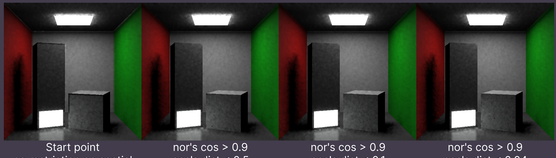
Start point
no restriction on spatial
average join: 14521.27
100%

nor's cos > 0.9
pos's dist < 0.5
average join: 14092.63
97.04%

nor's cos > 0.9
pos's dist < 0.1
average join: 13095.53
90.18%

nor's cos > 0.9
pos's dist < 0.04
average join: 5553.23
38.24%

Radius 2.5 - Area 19.625 reservoirs



Start point
no restriction on spatial
average join: 15011.42
100%

nor's cos > 0.9
pos's dist < 0.5
average join: 14743.55
98.21%

nor's cos > 0.9
pos's dist < 0.1
average join: 14698.51
97.91%

nor's cos > 0.9
pos's dist < 0.04
average join: 11608.62
77.33%

Ignoring GGX half's cos and gradually concentrating area results in better joining percentage, better specular and better shadows but it gradually adds fireflies. Oh boy, sure it would be good to join visibility separatelly :)
Reducing the sampling radius/count based on depth wasn't considered due to a small depth range in test scene.